Biospectra : Vol.9(1), March, 2014. pp 79-82
An International Biannual Refereed Journal of Life Sciences

Animal Sciences

ISSN : 0973-7057

Abundance and diversity of *Odontotermes spp.* on the bark of mango tree in Ranchi

Anjana Verma,[a] Naresh Kumar Mahto,[b]* Chandrama Singh[c] & M.Prasad[b]

[a]Department of Zoology, Y.S.College, Ranchi University, Ranchi

*[b]Dept. of Zoology, Ranchi University, Ranchi, Jharkhand, India.

[c]Department of Zoology, KTCT College, Raxaul.

Received 12th January, 2014; Revised 15th February, 2014

Abstract :*Odontotermes spp.* are one of the most notorious wood termite which has been sampled in amazing abundance from the bark of mango trees lower trunk of Ranchi city. The termites are not easily visible from the outer surface of the soil sheet which remains spread like soil coating on the bark of the tree giving a horrifying picture.

When the soil sheet is punctured or removed in field sampling from the bark of the tree, a large number of tinny grubs being the larval form of two different species identified as *Odontotermes obesus* Rambur and *Odontotermes parvidens* Holmgren emerged out from the hidden soil chambers or tunnel collectively known as bark termitorium. On the basis of numerical counting of the sampled termite species during rainy season, the abundance proportion of these two species has been calculated to the tune of 79.33% for *O.obesus* and 20.67% for *O. parvidens* respectively in Zone I (East) whereas it was 78.41% and 21.36% in Zone II (South) of Ranchi. From this statistical analysis it can be inferred that *O. obesus* is dominant over *O.parvidens*. The larvae also differ in their shape and size and biomass organization, the dominant species being smaller and light weight while recessive one larger and heavy weight. Further statistical indices related to these termite species diversity has also been discussed in the paper.

Keywords: *Odontotermes spp.,* Mango bark termite, Numerical dominance, Biomass, Organization, Abundance and diversity.

INTRODUCTION

Termites have an important economic role in economic entomology, with the cost of damage to the buildings, furniture especially in developed countries in America and Asia, amounting to many millions of pounds.

They are also called "white ants" as misnomer and are social insects of order Isoptera having a well developed caste system, with the presence of soldiers workers and reproductives.[1] Apart from their intrinsic biological interest, they are of great economic importance to mankind because they cause enormous damage running into billions of rupees annually to agriculture, horticulture, forestry plantations, stored products, ordnance depots, archives, libraries, all wood-work in buildings, wooden bridges, air fields, plywood, particle boards, fibre boards and plastics.[2] Damage to houses by termites can, in some countries, exceed that caused by natural disasters and fires in a single year. In developing countries they have even more impact, destroying local huts and crops of poor subsistence farmers. Villages in India and Egypt have been destroyed by termites and the inhabitants forced to move to other areas. In Asia, ancient temples have also been attacked.[3]

The present study has been done in the mango grooves of two specific zones East and South within the radius of 5 Km of Ranchi city. District Ranchi lies from 23.45^0 north latitudes and from 85.30^0 east longitudes. Its climate is very pleasant, dry in summer and cold and dry in winter. The summer season starts in April and continues up to June. The hottest months are May and June whereas the coldest months are December, January and February.

*Correspondent author :
Phone : 09835310072
E-mail : kumarnaresh34@yahoo.co.in

2 species of termites have been recorded from two different ecological zones (East and South) of Ranchi city, but there is paucity of information regarding their abundance and foraging activity in different climatic conditions.

Based on this consideration, termite population density in terms of relative abundance and diversity in the city of Ranchi was studied for comparison with other ecological zones of the area.

MATERIALS AND METHODS

Sampling of termites attacking mango trees

For taking samples of termites attacking trees, a mango groove plot of 100 x 100 m was selected in each zone (East and South) of Ranchi city. Nearly 50-120 mango plants existing in the 100x100m plot were examined at different localities of the zone to record the number of trees harbouring termite colonies. Collection was made from infested mango trees. Each infested mango tree was considered as quadrat for sampling.

In the identified plots of mango groove it was observed that some other trees like Acacia and Seesam were also planted at the fringe margins. Surprisingly these plants were also found to be infested by the termites and in the continuation of the study, samples of termites have also been collected and preserved.

Preservation of sampled specimens

Few representatives of the sampled individuals having morphological differences were preserved in 70% alcohol after killing them in the methyl acetate/ benzene jars.

Identification of the species

It was done by the taxonomic key of Imms[4] and confirmed by ZSI, Kolkata

Statistical Quantification of relative abundance and species diversity indices:

Statistical analysis of numerical data with respect to the sampling of termites collected during the period in these two zones (East and South) of Ranchi city had been done by using following statistical indices within the habitat.[5]

a) Relative abundance (RA) = ni/N x 100

Where,

ni = number of individuals of a species.

N = number of individuals of total number of species

b) Shannon Weiner Diversity index[6]: -

$$H = - \sum pi.\log pi$$

Where,

H = – diversity,

Pi = mean of individual species procured by the formula n/N, and log pi is the log product of pi (mean of individual species).

logpi = log value conversion of pi

RESULTS AND DISCUSSION

By the rigorous sampling of termite population from the soil sheet present on the bark of the mango tree trunks, only two varieties of termite species have been recorded with their distinctly different population density as furnished in table 1 and graph 1. The data of relative abundance of these two species clearly reflects the dominance of *Odontotermis obesus* over the other species *O.parvidens* the S-W diversity value of these two species has also been calculated and found to be ..... .(Table 1, Graph 1)

The second recessive species *O.parvidens,* however, was previously not reported from this area and it is predominantly found in Punjab. The possible cause of migration of this species may be the similarity of mango plantation in these two states.[7]

Termite species foraging on other trees

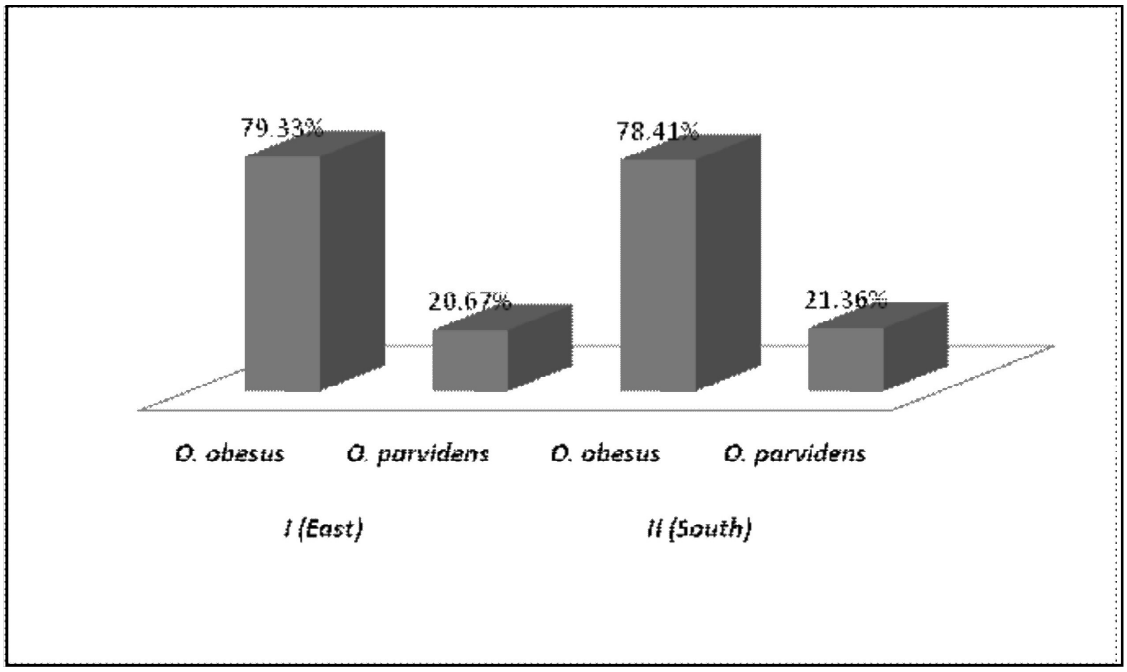
In addition to mango trees, the other trees like *Acacia modesta* and *Dalbergia sissoo* existing in the sampling area were also found to be infested by the termite species apparently different from the above two species. Taxonomically, they seem to have different affiliation to separate species, genera and family under further investigation. However, *Acacia modesta* has been found to be heavily infested by termite species than *Dalbergia sissoo* in these two sampling zones.

ACKNOWLEDGEMENT

The authors are immensely thankful to the mango groove farmers of Ranchi East (Tati Silwai) and South (Namkom) for their generous permission to enter into their farm for sampling of termites. Special thanks are being extended to ZSI, Kolkata for taxonomic identification of the sampled species.

Table 1: Relative Abundance and S-W diversity of termite species sampled from mango bark from Ranchi East and South mango grooves.

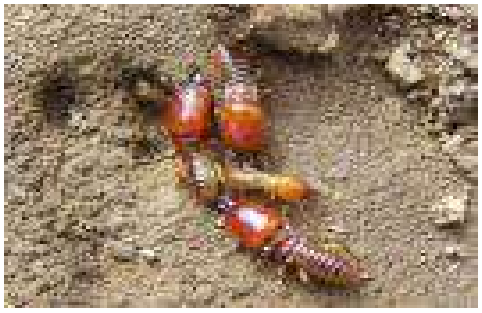
Zone	No. of plant	Sl. no.	Species of termites	No. of indiviuals per squ. m	S-W indices				
					RA = ni/ N × 100	pi=ni/N	Log pi	pi x log pi	H
I (East)	15	1	*O. obesus*	1785/ 2250	79.33%	0.7933	-0.10056	-0.07978	0.44732
		2	*O. parvidens*	465/ 2250	20.67%	0.2067	-0.68466	-0.14152	
II (South)	20	1	*O. obesus*	1725/ 2200	78.41%	0.7841	-0.10563	-0.08282	
		2	*O. parvidens*	470/ 2200	21.36%	0.2136	-0.6704	-0.1432	



Grpah 1: Relative Abundance of termite species sampled from mango bark from Ranchi East and South mango grooves.



Odontotermes obesus



Odontotermes parvidens

Fig.1: Sampled and identified species of termite from the mango bark of Ranchi East and South.



Fig.1: Infected mango tree trunk and bark of Ranchi East and South as the sampling sites of termites .

REFERENCES

1. **Tembhare, D.B. 2004.** Basic Entomology, Himalayan Publisher, Mumbai.
2. **Roonwal, M.L. 1979.** Termite life and Termite control in tropical South Asia. Scientific Publishers, Jodhpur.
3. **Pearce, M.J. 1997**. "Termites Biology and Pest Management" CAB International, New York.
4. **Richards, O.W. and Davies, R.G.. 1993.** Imms' general textbook of Entomology, Vol.2. B.I.Publications Pvt. Ltd. New Delhi.
5. **Stilling, P. 2002.** *Ecology* : Theories and applications. 4th Edn EEE , Prentice Hall of India, New Delhi.
6. **Shannon, C. E. and Weaner, W., 1949.** *The Mathmatical of Communication*. Urbana, Illinois : University of Illinois.
7. **Akhtar, M.S. and Sarwar, M. 1997**. "Termite population, diversity and damage in wheat fields of Bahawalpur division" *Proc. PakistanCongr. Zool*., 17, 153-163.

* * *